

Automated Sorting and Shredding System using Arduino Microcontrollers

Rennan M. Alabado

Department of Electronics Engineering
Caraga State University
Butuan City, Philippines
rmlabado@carsu.edu.ph

Jorich S. Loquero

Department of Electronics Engineering
Caraga State University
Butuan City, Philippines
jsloquero@carsu.edu.ph

Alexander P. Demetillo

Department of Electronics Engineering
Caraga State University
Butuan City, Philippines
atdemetillo@carsu.edu.ph

Lovely Mae Dagsa

Department of Electronics Engineering
Caraga State University
Butuan City, Philippines
lmdagsa@carsu.edu.ph

Jeffrey T. Dellosa ^{1,2}

Department of Electronics Engineering
¹Caraga State University
Butuan City, Philippines.
²MREC, Ateneo de Davao University,
Davao City, Philippines
jtdellosa@carsu.edu.ph

Ronieto N. Mendoza

Department of Electronics Engineering
Caraga State University
Butuan City, Philippines
rnmendoza@carsu.edu.ph

Abstract— The increasing demand for paper products, despite the push for digitalization, has led to significant environmental concerns, particularly in office settings where paper waste is prevalent. Manual sorting of paper waste is time-consuming, inefficient, and prone to errors, leading to substantial waste of reusable resources. This study develops an Automated Paper Sorter and Shredder to address these issues by efficiently sorting office paper based on its print status (printed on both sides, one side, or none) and shredding fully used papers to reduce waste volume. The machine integrates a smart paper feeder, color sensors for print detection, and a sorting mechanism with controlled gates and servo motors. Testing demonstrated a high detection accuracy of 96-97%, confirming the reliability of the system. Statistical analysis using SPSS software provided a confidence interval for detection errors, indicating consistent performance. The study concludes that the Automated Paper Sorter and Shredder can significantly enhance office paper recycling efforts, promote sustainable practices, and reduce the environmental impact of paper waste. Future improvements are recommended to expand the machine's capabilities and further optimize its performance.

Keywords— Automated paper sorter, shredding, office waste management, print detection, color sensors, recycling, sustainability, environmental impact, Arduino, paper recycling automation.

I. INTRODUCTION

A. Background

The paper industry, despite the modern push towards digitalization and paperless offices, continues to thrive and grow, presenting a paradox that underpins much of the global environmental discourse [1]-[5]. The Environmental Paper Network's report, "The State of the Global Paper Industry," highlighted that paper production has exceeded 400 million tons annually [6]. Over the past four decades, paper consumption has surged by 400%, with a significant 126% increase in the last twenty years alone [7]. By 2023, the global consumption of graphic paper is projected to reach 104 million tons [4]. The printing and writing sector accounts for a substantial portion of this consumption, with office papers forming a significant segment [8].

The paper industry is not only vast but also environmentally taxing [9]. Often dubbed the "world's biggest polluter," the industry's environmental impact starts from the raw material collection stage and extends through production to waste management [10]. Paper production is responsible for 40% of the world's logging, leading to an annual forest loss of 7.6 million hectares while only 4.3 million hectares are replanted [11]. Moreover, the industry is the largest consumer of freshwater, with a single A4 bond paper requiring 20 liters of water [12]. The production process also discharges toxic chemicals, contributing to water pollution, and emits significant amounts of carbon dioxide (CO₂), correlating with increased paper consumption [13]. Additionally, the paper industry consumes 4% of the world's energy, ranking as the fifth-largest industrial user [14].

The post-consumption phase further exacerbates the problem. In the United States, paper constitutes the highest percentage of solid waste, occupying 27% of landfills in 2013 and 25.9% in 2015, higher than plastics, which account for around 13% [15]. In the Philippines, paper is a major component of biodegradable waste, comprising 52% of solid waste [16]. Despite these figures, the recycling rate for printing and writing paper remains dismally low, with a global average recycled content of only 8% [17].

Given these challenges, there is a pressing need for innovative solutions to reduce paper waste and enhance recycling efforts. Various organizations and countries have already begun initiatives to minimize paper usage. For instance, Vodafone reduced its paper waste by 26.5 million sheets over three years through monitoring and changing printing behaviors. Similarly, the Royal Society for the Protection of Birds in Europe and the Dubai government significantly cut their paper consumption through systematic reviews and policy changes [1].

The manual sorting of office paper for recycling is time-consuming and inefficient, often leading to mismanagement of resources and increased waste. With growing efforts to promote double-sided printing to conserve paper, there is a critical need for an automated mechanism that can accurately sort paper based on its printing status (printed on both sides, one side, or none). Additionally, shredding papers printed on both sides can significantly reduce the volume of waste, making disposal more efficient.

B. Objectives

This study proposes the development of a 2-in-1 machine designed to automate the sorting and shredding of office paper. The machine will identify whether a paper is printed on both sides, one side, or not at all, and sort them accordingly. Papers printed on both sides will be shredded to save space, while those printed on one side or unprinted will be separated for potential reuse. The machine will feature a smart paper feeder, a print detection system using color sensors, and a sorting mechanism with controlled gates and servo motors. This automated solution aims to maximize the reusability of paper, reduce manual sorting efforts, and enhance the overall efficiency of paper recycling processes.

C. Significance

The significance of this study lies in its potential to address multiple environmental and operational challenges associated with paper waste. By automating the sorting process, the proposed machine will save time and reduce the labor-intensive nature of manual sorting. Economically, it can be implemented in both government and private offices to manage waste paper more effectively, promoting double-sided printing and reducing the consumption of new paper resources.

Environmentally, the machine supports sustainability by ensuring that paper is utilized to its maximum potential before being disposed of. By promoting double-sided printing and efficient sorting, the machine can help decrease the demand for raw materials, reduce energy consumption, and minimize the environmental footprint of paper production. The shredding feature further contributes to waste management by compacting paper waste, making disposal more manageable and reducing landfill space.

D. Sustainable Development Goals (SDGs) Addressed

The Automated Paper Sorter and Shredder study addresses several Sustainable Development Goals (SDGs), including:

- Goal 9: Industry, Innovation, and Infrastructure - By developing innovative technology to automate paper waste management, the study promotes sustainable industrial processes and infrastructure improvements.
- Goal 11: Sustainable Cities and Communities - The machine supports sustainable waste management practices, contributing to the reduction of urban waste and promoting recycling within communities.
- Goal 12: Responsible Consumption and Production - The study directly addresses the need to reduce waste through efficient use of resources and sustainable management of paper products, aligning with responsible consumption and production practices.



- Goal 13: Climate Action - By promoting the reuse of paper and reducing the environmental impact of paper production and disposal, the project contributes to efforts to mitigate climate change and reduce carbon emissions.
- Goal 15 : Life on Land - Efficient paper recycling and waste reduction help decrease deforestation and the degradation of natural habitats, supporting the conservation and sustainable use of terrestrial ecosystems.



The broader impact of this study extends to raising awareness about sustainable paper consumption practices. It highlights the importance of efficient resource use and recycling, aligning with global efforts to combat environmental degradation and promote sustainable development. This research not only addresses immediate operational needs but also contributes to long-term environmental conservation goals, positioning it as a valuable addition to both academic discourse and practical application in waste management.

II. METHODOLOGY

This chapter focuses on the construction and implementation of the smart paper sorter and shredder, covering the system's design, mechanisms, and data acquisition methods. The machine is designed to sort and shred office paper based on whether it is printed on both sides, one side, or not at all. The process is divided into four main subsystems: paper feeding, print identification, paper sorting, and paper shredding.

A. System Design and Implementation

The overall process flow of the machine is depicted in a series of diagrams that illustrate both the specified and simplified processes. The machine operates through a repeated cycle where paper is loaded, identified, sorted, and finally shredded if necessary.

- Paper Feeding Mechanism: The machine is equipped with a mechanism designed to hold a stack of papers. A pick-up roller, positioned on top of the stack, pulls the paper into the machine. An infrared (IR) proximity sensor detects the presence of paper and monitors the lifting mechanism of the paper tray to ensure continuous feeding.
- Print Identification: This subsystem uses color sensors to identify whether a paper is printed on both sides, one side, or is unprinted. The color sensors are strategically placed at the ends of the paper path with borders separating them to enhance detection accuracy.
- Paper Sorting Mechanism: Based on the data from the color sensors, the sorting mechanism utilizes controlled gates to divert the paper into appropriate bins. To ensure accurate sorting, there is a calculated distance between the second sensor and the first gate. Papers printed on both sides are directed to the shredder before disposal in the designated bin, while those printed on one side or unprinted are sorted into separate bins for potential reuse.

- **Paper Shredding:** Papers identified as printed on both sides are shredded to reduce their volume before being deposited in the final bin. This shredding process helps manage waste efficiently by compacting the paper.

B. System Overview

The mechanical and control systems of the machine are integrated and managed by an Arduino microcontroller, which serves as the system's brain (see Fig. 1). The Arduino is connected to various input and output components that facilitate the machine's operation. Inputs include the color sensors for print detection, IR sensors for proximity sensing, and a pushbutton to initiate the sorting process.

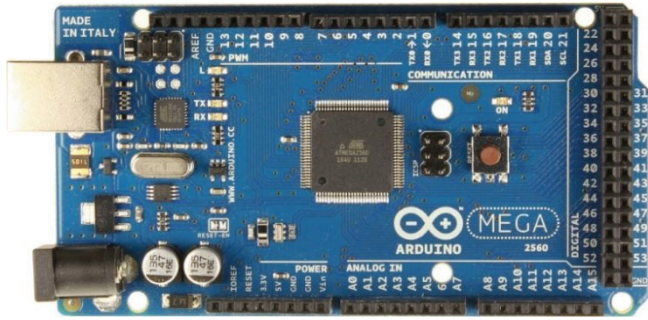


Fig. 1. The Arduino microcontroller, which serves as the system's brain, is connected to various input and output components that facilitate the machine's operation.

The outputs consist of multiple stepper motors and servo motors. Stepper motors drive the paper feeding mechanism, conveyance rollers, and the shredding unit, while servo motors control the diversion gates during the sorting process. The Arduino microcontroller is powered by a 5V source, with additional power sources for the stepper motors due to their higher power consumption.

- **Paper Feeding:** The paper feeding system includes an LED-LDR (Light Dependent Resistor) circuit to detect paper presence, stepper motors for lifting the paper stack, and a pick-up roller motor. The IR proximity sensor maintains the correct distance between the pick-up roller and the top of the paper stack, ensuring consistent feeding.
- **Print Detection:** The print detection system uses four color sensors (Fig. 2) and two LED-LDR circuits to identify the paper's print status. The sensors are enclosed to eliminate ambient light interference, and the detected frequencies are mapped to RGB values to distinguish between printed and unprinted paper.

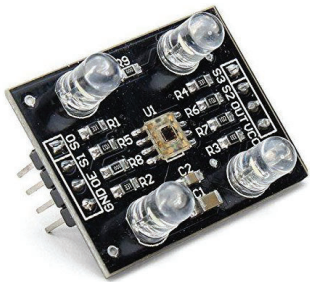


Fig. 2. The TCS230 Color Sensor module

- **Sorting Mechanism:** The sorting mechanism features servo motors and IR sensors to direct paper into the appropriate bins based on the print detection results. Each

bin entrance has a diversion gate controlled by the servo motors, and IR sensors detect the paper's entry to trigger gate retraction and the next paper feed.

- **Physical Construction:** The machine's physical structure is designed using SketchUp for 3D modeling. The main components include acrylic glass for the body, adhesives for assembly, and various mechanical parts like rollers, stepper motors, and a paper shredder.
- **Control System Components:** The control system includes an Arduino Mega microcontroller, TCS230 color sensor modules, ULN2003 and A4988 motor drivers, IR sensors, LDRs, and LEDs. These components work together to automate the paper sorting and shredding process efficiently.

Most stepper motors will operate only with the help of a driver module. It is because the controller module (In our case Arduino) will not be able to provide enough current from its I/O pins for the motor to operate. Such an external module, like the ULN2003 module and A4988, is used as a stepper module for 28BYJ-48 and NEMA 17 stepper motors. There are many types of driver modules, and the rating of one will change based on the type of motor used. The primary principle for all driver modules will be to source/sink enough current for the motor to operate. ULN2003 has the following specifications in Table 1.

TABLE I. ULN2003 STEPPER MOTOR SPECIFICATIONS

Minimum Operating Voltage	8V
Maximum Operating Voltage	35V
Continuous current per Phase	1A
Maximum Current per Phase	2A
Minimum Logic Voltage	3V
Maximum Logic Voltage	5.5V
Microstep Resolutions	1, 1/2, 1/4, 1/8, 1/16

C. Simple Statistical Analysis

The data gathered during testing is binary, indicating pass or fail based on the color sensor's detection accuracy. Random samples of office papers are tested, and the results are compared to visual assessments. The accuracy of detection is recorded, and statistical analysis is performed using SPSS Statistics software to calculate the confidence interval for detection errors.

In summary, the methodology encompasses the detailed design, construction, and testing of an automated paper sorter and shredder, integrating mechanical, electronic, and software components to achieve efficient and reliable paper waste management.

III. RESULTS AND DISCUSSIONS

The following results were gathered from the calibration and testing of the color sensor, as well as the installation of the servo motors covering the print identification phase and the sorting mechanism of the machine. Other subsystems, such as the paper feeding mechanism, which also involves paper conveyance, are not shown due to the inability to ship the necessary materials.

A. Construction of the Automated Paper Sorter

The construction of the machine involved the use of acrylic glass and adhesives. Using a laser cutter, the pieces for the surfaces were precisely cut and then assembled using adhesives. The construction process ensured that the machine's structure was stable and capable of supporting the various mechanical components required for its operation.

Figure 3 shows the actual smart paper sorter and shredder. The upper and lower guiding plates, where the paper is conveyed through rollers passing through the color sensing section, are depicted in Figure 4.



Fig. 3. The actual smart paper sorter and shredder.



Fig. 4. The color sensing section.

B. Calibration of Color Sensor

Calibration of the color sensor was a critical step to ensure accurate detection of paper prints. The sensor was placed in a 9" by 1" by 0.5" box without a bottom surface, with two square holes at the center of its top surface to fix the sensor's position. This setup was designed to remove ambient light, thereby providing accurate readings.

The sensor detects the frequency of the color reflected by the photodiodes, with a 20% scaled frequency in the Arduino. The detectable frequency was measured through several colors and mapped to levels of 0-255 corresponding to the RGB model. A reference color, black, was set to distinguish the length of the incoming paper.

The allowable whiteness of the paper was calibrated using two kinds of paper: one with no prints on both sides and one printed on both sides but still acceptable for printing. The frequency detected for these papers created the spectrum of colors that can be detected by the sensor.

C. Testing of Color Sensor

To test the calibrated color sensor, papers printed on one side were placed under the sensor. With the unprinted side facing the sensor, the output was "Pass," while with the printed side, the output was "Fail".

Due to limited resources, only three batches of 100 random samples of office papers were tested to evaluate the sensor's accuracy in detecting prints. The results are shown in Table 1. The accuracy of detection was recorded and compared to visual assessments. Accurate detection implies that both visual and sensor assessments were identical, whereas inaccurate detection indicates discrepancies between the two.

TABLE II. THE SENSOR TESTING ACCURACY RESULTS.

Batches	Results		
	<i>Accurate Detection</i>	<i>Inaccurate Detection</i>	<i>Total</i>
1	96	4	100
2	96	4	100
3	97	3	100

Table 2 presents the accuracy results of the sensor testing across three batches of 100 random samples of office papers. The table shows that in the first batch, 96 out of 100 samples were accurately detected, while 4 samples were inaccurately detected. The second batch also recorded 96 accurate detections and 4 inaccurate detections. In the third batch, the accuracy improved slightly, with 97 accurate detections and 3 inaccuracies. These results indicate a high level of consistency and reliability in the sensor's performance, with an average detection accuracy of approximately 96.33% across all batches. This high accuracy is crucial for the effectiveness of the automated paper sorting system, ensuring that papers are correctly identified and sorted based on their print status. The low number of inaccuracies highlights the robustness of the sensor calibration and the overall reliability of the machine in real-world applications. This performance is especially significant considering the constraints and limitations faced during the project's development and testing phases.

Table 3 shows the statistical analysis of the detection accuracy using SPSS Statistics software. With 95% confidence, the inaccurately detected samples are estimated to be between 2.2324 and 5.1009, approximately 2 to 6 samples per batch.

TABLE III. THE DESCRIPTIVE RESULTS FOR THE CONFIDENCE INTERVAL.

Total <i>Inaccurate 100.00</i>	Results		
		<i>Statistics</i>	<i>Std. Error</i>
	Mean	3.67	0.3333
	95% Confidence Interval for Mean	2.23 (lower bound) 5.10 (upper bound)	
	5% Trimmed Mean		
	Median	4.00	
	Variance	0.33	
	Std. Deviation	0.58	
	Minimum	3.00	
	Maximum	4.99	

Total	Results		
Inaccurate 100.00		Statistics	Std. Error
	Range	1.00	
	Skewness Kurtosis	-1.732	1.225

Table 3 presents the descriptive results for the confidence interval of the sensor testing accuracy. The table provides the upper and lower bounds of the confidence interval for the inaccurately detected samples across three batches of 100 random samples each. With a 95% confidence level, the inaccurately detected samples are estimated to fall between 2.2324 and 5.1009. This range indicates that, in most cases, the number of inaccuracies will lie between approximately 2 and 6 samples per batch.

The confidence interval reflects the consistency and reliability of the sensor's performance. Despite minor variations, the interval's narrow range demonstrates that the sensor maintains high accuracy across different batches. This statistical analysis reinforces the robustness of the automated paper sorting and shredding system, ensuring dependable sorting outcomes.

The confidence interval analysis is essential for validating the system's performance under various conditions, supporting the conclusion that the machine can consistently and accurately sort papers based on their print status. These results are crucial for potential users, assuring them of the system's reliability and effectiveness in real-world applications, thereby promoting its adoption for efficient and sustainable office paper waste management..

D. Construction and Assembly Challenges

One of the main challenges faced during the construction and assembly of the machine was the limitation of available materials. Despite these limitations, the machine was constructed using alternative materials where possible and adjustments were made to ensure functionality.

The use of acrylic glass and adhesives for the machine's body provided a lightweight yet durable structure. The laser-cut acrylic pieces were essential for precise fitting and assembly, which was critical for the machine's performance. The use of high-torque stepper motors and controlled gates ensured that the paper sorting and shredding processes were efficient and reliable.

E. Challenges Efficiency of the Sorting Mechanism

The sorting mechanism was designed to accurately direct papers into appropriate bins based on their print status. The use of servo motors for controlling the diversion gates proved to be effective in sorting papers into three categories: unprinted, printed on one side, and printed on both sides. The IR sensors played a crucial role in detecting the paper's entry into the bins, triggering the appropriate response from the servo motors.

The shredding mechanism for papers printed on both sides was also effective in reducing the volume of waste, making disposal more manageable. The integration of the shredding process into the sorting mechanism ensured that the machine could handle large volumes of paper efficiently.

F. Data Analysis and Accuracy

The data gathered from the testing phase was binary, indicating either pass or fail based on the sensor's detection

accuracy. The accuracy of the sensor was high, with an average detection accuracy of approximately 96-97% across the three batches tested. This high level of accuracy is critical for ensuring that the machine can reliably sort papers based on their print status.

The statistical analysis using SPSS Statistics software provided a confidence interval for the detection errors, indicating that the sensor's performance is consistent and reliable. The results demonstrate that the machine can effectively distinguish between printed and unprinted papers, ensuring that the sorting process is both accurate and efficient.

G. Implications for Office Paper Management

The automated paper sorter and shredder have significant implications for office paper management. By automating the sorting and shredding processes, the machine reduces the need for manual sorting, which is both time-consuming and prone to errors. The machine's ability to accurately identify and sort papers based on their print status ensures that resources are used efficiently, promoting the reuse of unprinted and single-sided papers.

Economically, the machine offers a cost-effective solution for managing office paper waste. By reducing the volume of paper waste through shredding and promoting the reuse of partially used papers, offices can lower their paper consumption and waste disposal costs. Environmentally, the machine supports sustainable paper management practices by minimizing waste and promoting recycling.

H. Future Enhancements

While the current design and implementation of the automated paper sorter and shredder have proven effective, there are several areas for future enhancements. One potential improvement is the integration of more advanced sensors that can detect additional properties of the paper, such as thickness and quality. This would allow for even more precise sorting and recycling processes.

Another area for enhancement is the expansion of the machine's capacity to handle different paper sizes and types. By incorporating adjustable settings and components, the machine could be adapted to sort a wider range of paper products, making it even more versatile and useful in various office environments.

Additionally, the incorporation of data logging and analysis features could provide valuable insights into paper usage and waste patterns. This data could be used to further optimize the machine's performance and support sustainability initiatives within organizations.

IV. SUMMARY

The Automated Paper Sorter and Shredder project was developed to address the inefficiencies and environmental impact associated with manual sorting of office paper waste. The study aimed to design and prototype a machine capable of identifying, sorting, and shredding office paper based on its print status (printed on both sides, one side, or not at all). This automated solution seeks to promote the reuse of partially used paper, reduce waste volume, and support sustainable paper management practices.

Key findings and achievements are provided as follows based on the development of this system:

1. System Design and Implementation:

The machine integrates four primary subsystems: paper feeding, print identification, paper sorting, and shredding. A robust mechanical structure was constructed using acrylic glass and adhesives, ensuring stability and precision in the machine's operations. The system is controlled by an Arduino microcontroller, which manages inputs from color sensors and IR sensors and outputs to stepper motors and servo motors.

2. Calibration and Testing:

The color sensor was calibrated to accurately detect whether a paper was printed on both sides, one side, or not at all. The machine's sorting mechanism was tested with three batches of 100 random samples of office papers, demonstrating a high detection accuracy of 96-97%. Statistical analysis confirmed the reliability of the sensor, with an estimated error rate of 2-6 samples per batch.

3. Efficiency and Performance:

The sorting mechanism effectively directed papers into appropriate bins, with shredded paper printed on both sides to reduce disposal volume. The machine's automation significantly reduces the time and labor required for manual paper sorting, enhancing operational efficiency.

4. Challenges and Limitations:

The material availability and project timeline, impacting the construction and testing phases. The machine was tested with constant paper size and quality, which may not fully represent real-world variability.

5. Implications and Future Enhancements:

The machine offers a cost-effective solution for managing office paper waste, promoting double-sided printing and reducing new paper consumption. Future enhancements could include advanced sensors for detecting additional paper properties, adjustable settings for different paper sizes, and data logging for usage analysis.

The *Automated Paper Sorter and Shredder* project successfully developed a functional prototype that addresses the critical need for efficient paper waste management in office environments. The machine's high accuracy in identifying and sorting papers, combined with its shredding capability, offers a comprehensive solution for reducing paper waste and supporting sustainability initiatives.

Future improvements and refinements can further enhance the machine's functionality and versatility, contributing to broader efforts to promote environmentally responsible practices in both public and private sectors.

V. RESULTS AND DISCUSSIONS

The *Automated Paper Sorter and Shredder* project successfully developed a prototype capable of automating the sorting and shredding of office paper based on its print status. This innovative machine addresses significant challenges associated with manual paper sorting, including time inefficiency and the potential for human error. By integrating mechanical, electronic, and software components, the machine demonstrated high accuracy in identifying and sorting papers, thereby promoting the reuse of partially used paper and reducing the volume of paper waste through shredding. The project's success underscores the feasibility

and effectiveness of automating office paper waste management processes. The machine's design not only enhances operational efficiency but also supports sustainability initiatives by encouraging responsible paper use and disposal practices.

ACKNOWLEDGMENT

We would like to express our deepest gratitude and sincere appreciation to everyone who supported and guided us throughout the development of this project. We are immensely grateful to our thesis adviser, Engr. Alexander P. Demetillo, for his invaluable guidance and expertise, and to the members of our advisory committee, including Jeffrey T. Delloso, Lovely Mae Dagsa, Lawrence P. Grana, Judyrose Q. Lamoste, Lloyd B. Lloren, Ronieto N. Mendoza, Isagani M. Roma Jr., Edgardo Ricardo B. Sajonia Jr., and Ydron Paul C. Amarga, for their constructive feedback and suggestions. We also extend our thanks to the Fablab Laboratory and the Center for Robotics Automation and Fabrication Technologies (CRAFT) for providing the necessary equipment and workspace.

REFERENCES

- [1] Environmental Paper Network (2018). The State of the Paper Industry 2018, Shifting Seas: New Challenges and Opportunities for Forests, People and the Climate
- [2] Ezeudu, Obiora, et.al. (2019). Sustainable Production and Consumption of Paper and Paper Products in Nigeria: A Review
- [3] Green America (2017) Paper Production and Consumption Facts. Retrieved from <https://www.greenamerica.org>
- [4] Business Wire (2019). Global Graphic Paper (Printing & Writing Newsprint) Market Insights, Trends and Forecast Report 2019-2023. Retrieved from Researchmarkets.com
- [5] Sopha, Bertha Maya (2013). Sustainable Paper Consumption: Exploring Behavioral Factor
- [6] Ezeudu, O. B., Agunwamba, J. C., Ezeasor, I. C., & Madu, C. N. (2019). Sustainable production and consumption of paper and paper products in Nigeria: A review. Resources, 8(1), 53.
- [7] Bajpai, P. (2016). Pulp and paper industry: energy conservation. Elsevier.
- [8] Bajpai, P. (2024). Recycling and deinking of recovered paper. Elsevier.
- [9] Xu, J. (1999). China's paper industry: growth and environmental policy during economic reform. Virginia Polytechnic Institute and State University.
- [10] Bendell, J. (2023). World Review. In Australasian Perspectives on Corporate Citizenship (pp. 4-18). Routledge.
- [11] Cheng, B., & Le Clue, S. (2010). Forestry in Asia-Issues for Responsible Investors. Responsible Research.
- [12] Esmaeeli, A., Sarrafzadeh, M. H., Zeighami, S., Kalantar, M., Bariki, S. G., Fallahi, A., ... & Ghaffari, S. B. (2023). A comprehensive review on pulp and paper industries wastewater treatment advances. Industrial & Engineering Chemistry Research, 62(21), 8119-8145.
- [13] Gupta, G. K., & Shukla, P. (2020). Insights into the resources generation from pulp and paper industry wastes: challenges, perspectives and innovations. Bioresource technology, 297, 122496.
- [14] Qian, K. (2021). Environmental Concerns of the Pulp and Paper Industry: Focusing on Household and Sanitary Paper Products. Decision Science for Future Earth: Theory and Practice, 181-195.
- [15] Mukherjee, C., Denney, J., Mbonimpa, E. G., Slagley, J., & Bhowmik, R. (2020). A review on municipal solid waste-to-energy trends in the USA. Renewable and Sustainable Energy Reviews, 119, 109512.
- [16] Pintor, L. L., & Ancheta, A. A. (2022). Meta-analysis of studies on solid waste management in the Philippines. In Effective Waste Management and Circular Economy (pp. 243-253). CRC Press.
- [17] Goldstein, J. L. (2020). Rethink the Bins: Your Guide to Smart Recycling and Less Household Waste. Bebo Press.